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SKILL WALL
A SmartBridge Project

Data Flow Diagram

Date	29 September 2026
Team ID	SWTID-2026-9550
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through the Script-Controlled ACL system between users, ACL processes, and the Institution Details data store. The diagram represents how user requests are checked against roles and field values before access is allowed or denied.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	A user or administrator outside the ACL process who sends an access request and receives the result.
Rectangle with numbered header	**Process**	An activity that evaluates the request, user role and field value, then produces an access decision.
Rectangle (solid fill)	**Data Store**	A place where Institution Details records and ACL-related data are held for later use.
Labelled arrow	**Data Flow**	Movement of user requests, record data and access decisions between entities, processes and the data store.

Example

The diagram below illustrates how a user request flows through role checking, field-value checking, record access and the final allow/deny response.

USER / ADMIN	**1. ACCESS REQUEST** Receive record request and identify user role	**2. ACL EVALUATION** Check operation, role and Branch condition	**INSTITUTION DETAILS** Student records u_institution_details
Access request record + operation	User role (bb1–bb4)	Branch / ACL decision	Requested record data

USER / ADMIN Receives result	**3. ACCESS DECISION** Allow or deny based on ACL evaluation	**RECORD OPERATION** Read / Create / Write / Delete when permitted	**ACL RULES** Read: bb1 + Branch condition Create: bb2 Write: bb3 Delete: bb4

Data flow: User/Admin → Access Request → ACL Evaluation → Institution Details → Access Decision → Allowed record operation or denied access.